

Please check the examination details below before entering your candidate information

Candidate surname

Other names

**Pearson Edexcel
Level 3 GCE**

Centre Number

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Candidate Number

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Mock Paper Set 1

Paper Reference **8MA0/22**

Mathematics

**Advanced Subsidiary
Paper 22: Mechanics**

You must have:

Mathematical Formulae and Statistical Tables (Green), calculator

Total Marks

**Candidates may use any calculator allowed by Pearson regulations.
Calculators must not have the facility for symbolic algebra manipulation,
differentiation and integration, or have retrievable mathematical formulae
stored in them.**

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear.
Answers without working may not gain full credit.
- Unless otherwise indicated, wherever a value of g is required, take $g = 9.8 \text{ m s}^{-2}$ and give your answer to either 2 significant figures or 3 significant figures.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- The total mark for this part of the examination is 30. There are 4 questions.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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1.

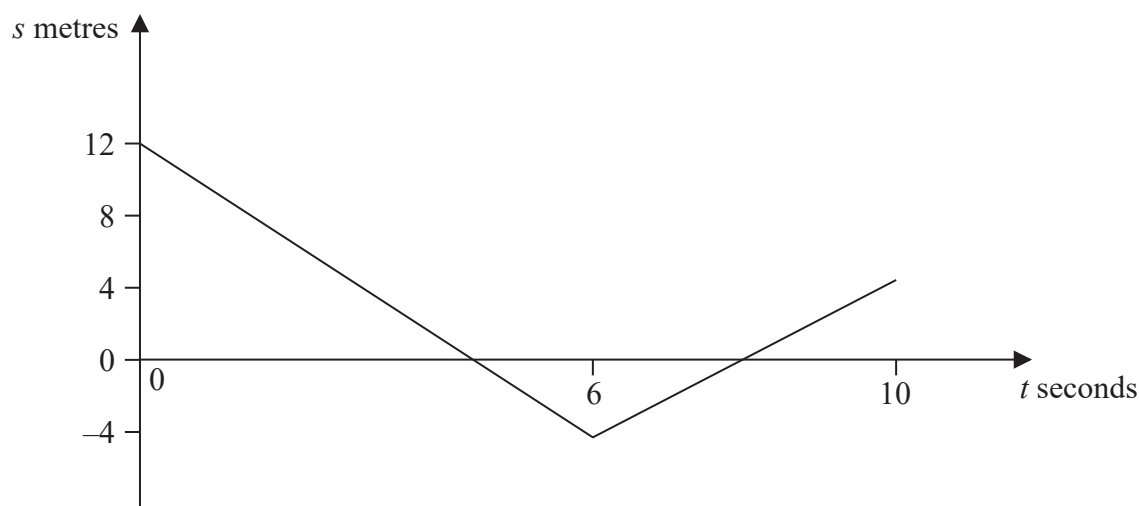


Figure 1

A toy car moves along a straight line.

The line contains the fixed point O

The toy car is modelled as a particle.

Figure 1 shows the **displacement-time** graph for a model of the motion of the toy car, where s metres is the displacement of the toy car from O at time t seconds.

Using the model,

- (a) find the speed of the toy car in the interval $0 \leq t \leq 6$ (1)
- (b) find the average speed of the toy car in the interval $0 \leq t \leq 10$ (2)
- (c) sketch the **velocity-time** graph for the motion of the toy car in the interval $0 \leq t \leq 10$ (2)
- (d) (i) Referring to your sketch, or otherwise, describe what happens to the motion of the toy car at the instant when $t = 6$
- (ii) Explain why your answer to (i) represents a limitation of the model of the motion of the toy car at $t = 6$ (2)



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Question 1 continued

Lined area for writing the answer to Question 1.



Question 1 continued

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Question 1 continued

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(Total for Question 1 is 7 marks)



2.

[In this question $\mathbf{i}$ and $\mathbf{j}$ are horizontal unit vectors.]

A particle P of mass 5 kg is moving on a smooth horizontal plane.

The particle P is moving under the action of two forces $(\mathbf{i} + 2\mathbf{j})\text{N}$ and $(-c\mathbf{i} - 10\mathbf{j})\text{N}$, where c is a positive constant.

The magnitude of the acceleration of P is 2 m s^{-2}

Find the value of c

(5)



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Question 2 continued

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(Total for Question 2 is 5 marks)



3. A fixed point O lies on a straight line.

A particle P moves along this line.

At time $t = 0$, particle P passes through O

At time t seconds, $t \geq 0$, the velocity, $v \text{ m s}^{-1}$, of P is

$$v = 6 + 4t^{\frac{3}{2}} - 3t^2$$

- (a) Find the acceleration of P at time t seconds.

(2)

- (b) Find the distance of P from O at the instant when $t > 0$ and the acceleration of P is zero.

(6)



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Question 3 continued

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(Total for Question 3 is 8 marks)



4.

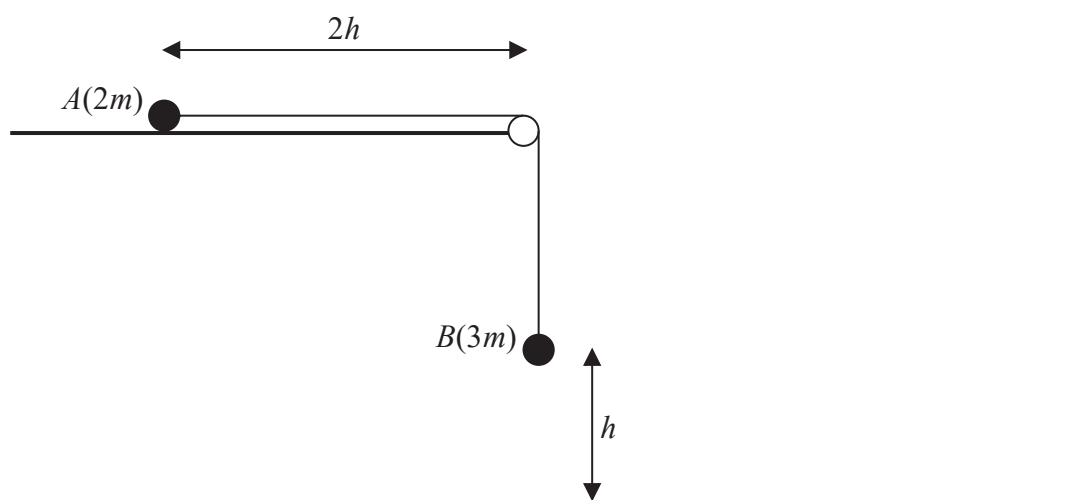


Figure 2

A small smooth light pulley is fixed to the edge of a rough horizontal table.

Two particles A and B are connected by a string which passes over the pulley.

- Particle A has mass $2m$
- Particle B has mass $3m$
- Particle A is held at rest on the rough table at a distance of $2h$ from the pulley
- Particle B hangs freely below the pulley, with the string taut, at a height h above a horizontal floor, as shown in Figure 2

Particle A is released from rest and begins to move along the table.

In an initial model,

- the string is modelled as being light and inextensible
- the resistance to motion that the table exerts on A is modelled as a constant force of magnitude $\frac{1}{2}mg$
- air resistance is modelled as being negligible

This model predicts that in the subsequent motion, B first strikes the floor with speed V

Using this initial model,

- write down an equation of motion for A (2)
- find V in terms of g and h (6)

Question 4 continues on page 11



This improved model predicts that in the subsequent motion, B first strikes the floor with speed W

- (c) State which speed is greater, V or W , explaining your answer. (1)
- (d) State one further improvement to the model that would make the model more realistic. (1)

Question 4 continued

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(Total for Question 4 is 10 marks)

TOTAL FOR MECHANICS IS 30 MARKS

